A LOCAL COMPLETE INTERSECTION MORPHISM IS LOCALLY OF FINITE PRESENTATION, PSEUDO-COHERENT AND PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 18160-18175, immediately following the statement at 18149-18158.

Editorial note. Same-name twin check: the identical author label in spaces-more-morphisms (statement 11207-11216) has an omitted proof and is therefore not counted; this card is the schemes version with the complete author proof.

Editorial note. Three chapter lemmas are cited by \ref as navigation pointers; the reduction chain that uses them is inside the counted span.

Allowed standard prerequisites (editorial).

- A Koszul-regular immersion is perfect. chapter lemma lemma-regular-immersion-perfect, cited at proof line 18169. Supplies perfectness of the closed immersion i in the local factorisation of the morphism.
- A flat morphism of finite presentation is perfect. chapter lemma lemma-flat-finite-presentation-perfect, cited at proof line 18172. Supplies perfectness of the smooth morphism pi in the factorisation.
- Composition of perfect morphisms is perfect. chapter lemma lemma-composition-perfect, cited at proof line 18174. Combines the two pieces into perfectness of the composed morphism.
- Perfect ring maps are pseudo-coherent and pseudo-coherent ring maps are of finite presentation. used at proof lines 18161-18165. Derives claims (1) and (2) from claim (3), reducing the whole lemma to perfectness.
- Local nature of perfectness. used at proof line 18165. Allows the defining local factorisation into a smooth morphism followed by a Koszul-regular immersion to be assumed.

Source reference. Original lemma 0689: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17760–17760. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 068A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17783–17783. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 068C: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17814–17814. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a local complete intersection morphism. Then*

(1) *f is locally of finite presentation,*
(2) *f is pseudo-coherent, and*
(3) *f is perfect.*

Proof. Since a perfect morphism is pseudo-coherent (because a perfect ring map is pseudo-coherent) and a pseudo-coherent morphism is locally of finite presentation (because a pseudo-coherent ring map is of finite presentation) it suffices to prove the last statement. Being perfect is a local property, hence we may assume that f factors as $\pi \circ i$ where π is smooth and i is a Koszul-regular immersion. A Koszul-regular immersion is perfect, see Lemma 068C. A smooth morphism is perfect as it is flat and locally of finite presentation, see Lemma 068A. Finally a composition of perfect morphisms is perfect, see Lemma 0689. □